

Case Study 5

Maximizing the Income for Wind Power Plant Integrated with a Battery Energy Storage System Using Dynamic Programming

This case study is based on the following research paper:

M. Khalid, A. V. Savkin, and V. G. Agelidis, "Maximizing the income for wind power plant integrated with a battery energy storage system using dynamic programming," in *Proceedings of the 10th Asian Control Conference 2015 (ASCC15)*, Kota Kinabalu, Malaysia, June 2015.

Introduction and Motivation

- Renewable Energy (RE) like Wind and Solar is in highest demand nowadays all over the world,
- RE is generally considered as a valuable supplement to conventional energy resources.



Figure: Wind Farm



Figure: Solar PV Pannels

Drawbacks and Problems

- We target **Wind Energy** and address the associated issues.
- The efficient and cost effective integration of Wind Energy into electricity grid has become the greatest challenge due to following inherent problems:
 - it is a highly fluctuating resource,
 - it is not easily predictable and controllable,
 - it is subjected to variability problems,
 - larger levels of such intermittent generation may have implications for electricity network security and stability.

A General Solution - Battery Energy Storage

- Battery Energy Storage System (BESS) is highly utilized in many RE applications such as to suppress the wind power fluctuations, as a backup power supply, frequency control, and power dispatch.

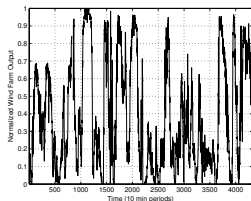


Figure: Wind Power Profile

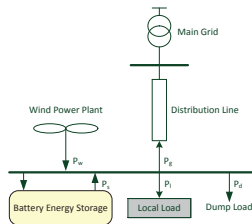


Figure: Wind Power with Storage

Battery Energy Storage - Challenges

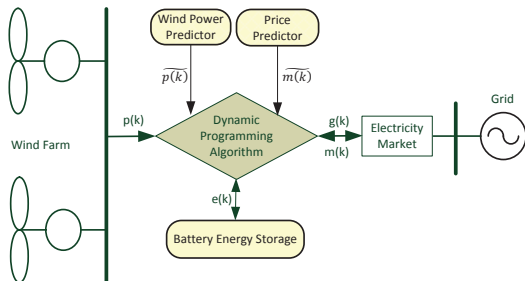
- Although BESS technology plays an important role to increase the penetration of wind power into the electricity grid but there are significant associated challenges. For example:
 - BESS is still expensive,
 - a large BESS is required for high penetration of wind power and normally BESS is over-dimensioned,
 - a complicated control circuitry is required for an efficient operation of BESS,
 - **Capacity Optimization** and accurate sizing is vital for cost effective solutions.

The Proposed Methodology

- We propose **Constraint-based Charging/Discharging** of the BESS coupled with the associated forecasting systems:
 - The proposed strategy originates from the fact that wind power and electricity market predictions play an important role in developing an effective wind power dispatch strategy.
 - Furthermore, restricting the charging/discharging of the BESS within the given practical upper and lower limits will enhance the battery lifetime.
 - Finally, developing an effective wind power dispatch strategy particularly with an optimal capacity of the BESS will maximize the income for wind power plant.

The Proposed Methodology

- The block diagram of the proposed methodology



Problem Formulation - The System

We consider a grid-connected wind farm integrated with a BESS. Let $p(k)$ denote the energy provided by the wind farm at time k , and $g(k)$ denote the amount of energy sold to the market at time k . Clearly, the following constraint must be satisfied:

$$g(k) \geq 0, \quad \forall k \geq 0 \quad (1)$$

Furthermore, let $e(k)$ denote the energy difference

$$e(k) = p(k) - g(k) \quad (2)$$

which is sent to the battery energy storage assuming that the conversion stages are lossless. If $e(k) > 0$ the BESS is charged, if $e(k) < 0$ the BESS is discharged.

Problem Formulation - The System

Moreover, let $m(k)$ be the market energy price at time k . Therefore, the amount of money received at time k is given by $m(k)g(k)$.

In this work, we assume that the BESS has capacity $c > 0$. Let $x(k)$ denote the amount of energy at time k in the battery. Therefore, $x(k)$ satisfies the following discrete-time dynamic model:

$$x(k+1) = x(k) + p(k) - g(k) \quad (3)$$

Additionally, the battery has a maximum charge rate $r_c > 0$ and a maximum discharge rate $r_d > 0$, i.e., the state of the battery satisfies the constraint

$$-r_d \leq x(k+1) - x(k) \leq r_c, \quad \forall k \geq 0 \quad (4)$$

Problem Formulation - The System

Furthermore, to prolong the battery life and reduce the cost of the BESS, we require the following constraint:

Let $0 < \alpha_m < \alpha_M < 1$ be given constants. Then

$$\alpha_m c \leq x(k) \leq \alpha_M c, \quad \forall k \geq 0 \quad (5)$$

Where α_m and α_M are the lower and upper limits of BESS respectively (e.g., 20% - 80% operational band).

Thus we consider system (3) as a control system with $g(k)$ as the control input which has to satisfy constraints (1), (4) and (5).

Here we use the concept of Model Predictive Control (MPC) and propose to solve the problem using Dynamic Programming (DP) algorithm.

Problem Formulation - Objective Function

To derive an optimal wind power dispatch strategy, we first consider the economic cost function associated with the system as:

$$V_N(x, \vec{g}(k)) = \sum_{j=0}^{N-1} \hat{m}(j) \hat{g}(j) \quad (6)$$

where $N > 0$ is the prediction horizon and $\hat{m}(j)$ stands for the predicted future value of the energy price, where $\hat{m}(0) = m(k)$, i.e., the current energy price is known. Additionally, vector $\vec{g}(k)$ contains the tentative control actions over the prediction horizon, i.e.,

$$\vec{g}(k) = \{\hat{g}(0), \hat{g}(1), \dots, \hat{g}(N-1)\} \quad (7)$$

Problem Formulation - System Constraints

Consequently, the MPC optimization of interest for the current states $x(k) = x$, is given by

$$\mathbb{P}_N(x) : \quad V_N^{op}(x) = \max_{\vec{g}(k)} \{ V_N(x, \vec{g}(k)) \} \quad (8)$$

$$\text{subject to:} \quad \hat{x}(j+1) = \hat{x}(k) + \hat{p}(j) - \hat{g}(j) \quad (9)$$

$$\hat{g}(j) \geq 0 \quad (10)$$

$$\alpha_m c \leq \hat{x}(j) \leq \alpha_M c \quad (11)$$

$$-r_d \leq \hat{x}(j+1) - \hat{x}(j) \leq r_c \quad (12)$$

for all $j \in \{0, \dots, N-1\}$, where $\hat{x}(0) = x(k)$ and $\hat{p}(0) = p(k)$, i.e., the current battery state and wind power are known.

Problem Formulation - Optimal Control

Hence the optimal input sequence, $\vec{g}^{op}(k)$, is the one which maximizes the cost function in presence of system constraints, i.e.,

$$\vec{g}^{op}(k) \triangleq \arg \left\{ \max_{\vec{g}(k)} V_N(x, \vec{g}(k)) \right\} \quad (13)$$

The resulting optimal solution is so-called the input control sequence of the form:

$$\vec{g}^{op}(k) = \{g^{op}(0), g^{op}(1), \dots, g^{op}(N-1)\} \quad (14)$$

Problem Formulation - Receding Horizon

One key aspect of MPC is the receding or moving horizon principle which states that only the first element from the optimal input sequence in (14) is applied to the system. Then the optimization process is repeated at the next sampling instant using new predicted values for $m(\hat{k})$ and $p(\hat{k})$ to obtain a new optimal input sequence, $\vec{g}^{op}(k+1)$. Thus the solution of the optimal control problem $\mathbb{P}_N(x)$ yields the following control law

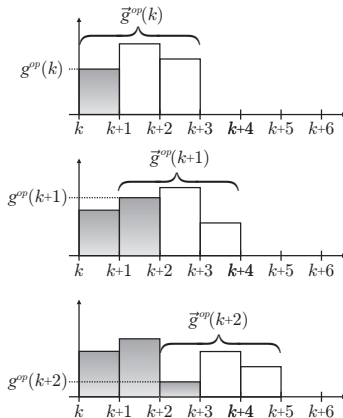
$$g^{op}(k) = g^{op}(0) \quad (15)$$

Therefore, the resulting MPC closed-loop system can be represented for each sampling instant as:

$$x(k+1) = x(k) + p(k) - g^{op}(k) \quad (16)$$

Problem Formulation - Receding Horizon

- The receding or moving horizon principle for $N = 3$



Problem Formulation - Dynamic Programming

As mentioned earlier, we propose to solve the given optimal control problem using the DP algorithm. The DP is an off-line optimization method that works backward starting from the final time and goes back to the initial time.

Therefore, by invoking the well known [Bellman's optimality principle](#), we immediately obtain the following algorithm for computing the optimal control sequence $\vec{g}^{op}(k)$.

$$\begin{aligned} V(\hat{x}, N) &= 0, \\ V(\hat{x}, j) &= \max_{\hat{g}(j)} \{ \hat{m}(j) \hat{g}(j) + V(\hat{x}(j) + \hat{p}(j) - \hat{g}(j), j+1) \}, \end{aligned} \quad (17)$$

for all $j \in \{N-1, N-2, \dots, 0\}$.

Problem Formulation - Dynamic Programming

Here (17) encompasses the optimal subproblems that need to be solved satisfying the original constraints (9)-(12) for each prediction step. To do so, we translate the state constraints (11) and (12) into input constraints by using the system dynamic constraint (9). Hence, it follows that

$$\begin{aligned}\hat{x}(j) + \hat{p}(j) - \alpha_M c &\leq \hat{g}(j) \leq \hat{x}(j) + \hat{p}(j) - \alpha_m c \\ \hat{p}(j) - r_c &\leq \hat{g}(j) \leq \hat{p}(j) + r_d\end{aligned}\tag{18}$$

Consequently, it is important to highlight that the resulting final solution (optimal control sequence $\vec{g}^{op}(k)$), lies at an extreme point of the set of admissible controls (upper and lower bounds). This form of policy is known as bang-bang control.

Problem Formulation - Remark

Remark

It is important to highlight that our proposed strategy relies on the predicted values of price $\hat{m}(k)$ and wind power $\hat{p}(k)$. Therefore we have analyzed the system with prediction error of up to 10% in both the predictors in each of our moving horizon window based on the fact that the root mean square of the short-term prediction error can be reduced to below than 10%.

Databases

- Wind was obtained data from [Roaring 40s Woolnorth Wind Farm](#) in Tasmania, Australia. Electricity market price data was obtained from Australian electricity market operator.



Figure: Geographical Location of Data Site

Simulation Results

- Typical Profiles

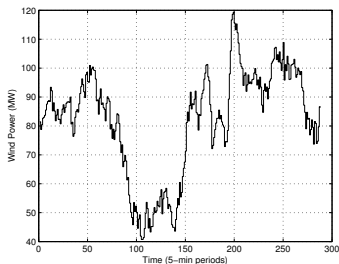


Figure: A typical wind power.

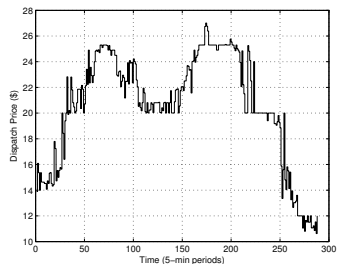


Figure: A typical market price.

Simulation Results

- Daily Income Improvement Over a Trivial Strategy

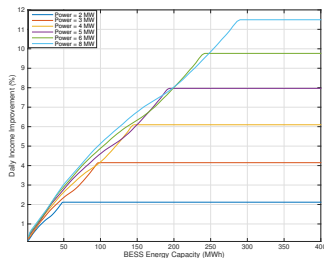


Figure: Percentage of Improvement for a few selected BESSs.

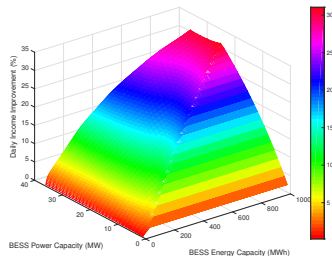


Figure: Percentage of Improvement for a broad range of BESSs.

Simulation Results

- Battery States During an Operational Cycle
($r_c^{max} = 36MW/5min = r_d^{max}$)

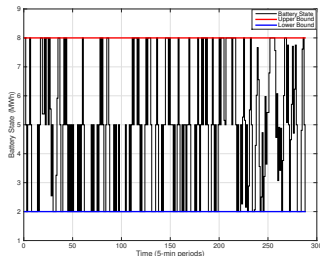


Figure: Higher power capacity with lower energy capacity.

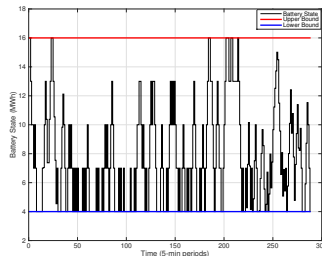
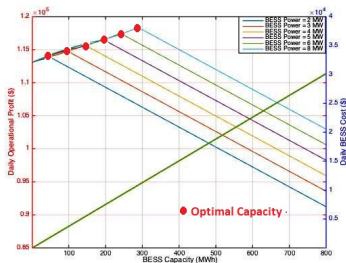


Figure: Higher power capacity with higher energy capacity.

Simulation Results

- Daily BESS cost (right y-axis) vs daily operational profit (left y-axis) for BESSs
- Theoretically, each optimal point refers to BESS Energy Capacity corresponding to selected power rating of the BESS



Concluding Remarks

- A Dynamic Programming algorithm has been used to develop an optimal wind power dispatch strategy based on the predicted values of wind power and market price
- The proposed strategy has been evaluated for different BESSs in terms of both energy and power capacities with variable energy pricing of the given electricity market satisfying the given operating constraints
- Finally, the optimal capacity of BESS has been determined for different power and energy ratings